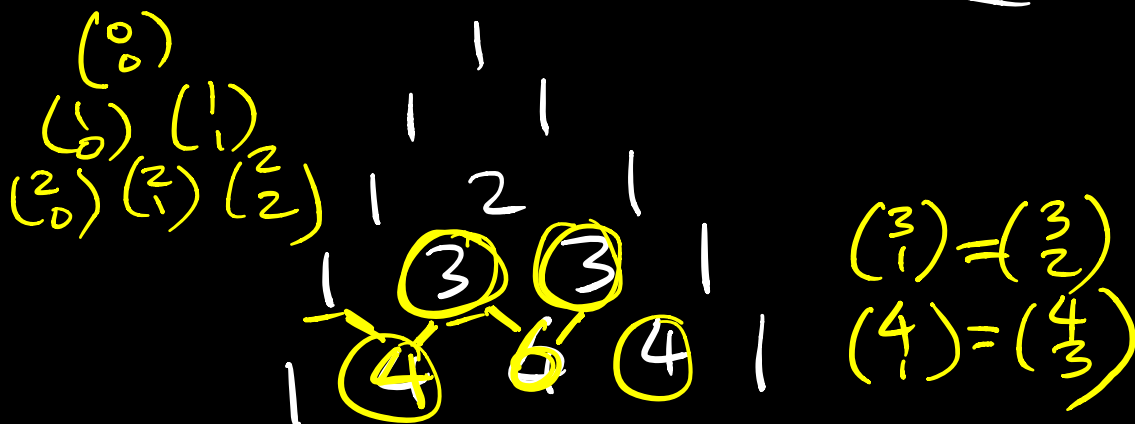


Lesson 13 More Combinatorial Proofs



Prop 1 $\binom{n}{k} = \binom{n}{n-k}$

Algebraic proof

$$\binom{n}{k} = \frac{n!}{(n-k)! k!} \quad \binom{n}{n-k} = \frac{n!}{\underbrace{(n-(n-k))!}_{k!} (n-k)!}$$

Combinatorial proof

To construct a k -element subset of an n -element set, we may choose k elements to include or we may choose $n-k$ elements to exclude.

Therefore there ^{are} $\binom{n}{k}$ such subsets or $\binom{n}{n-k}$ such subsets, and $\binom{n}{k} = \binom{n}{n-k}$.

Prop 2 $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$
Pascal's Identity
Combinatorial Proof

LHS: # of k-element subsets of
 $A = \{1, 2, \dots, n\}$

RHS: There are 2 types of such subsets:

(1) Subsets with "n": $\binom{n-1}{k-1}$
 $\{n, \underbrace{\dots}_{k-1}\}$

(2) Subsets without "n": $\binom{n-1}{k}$
 $\{\underbrace{\dots}_k\}$

Prop 3 $k \binom{n}{k} = n \binom{n-1}{k-1}$

Proof How many ways to choose
 from n people a team of k members
 with a captain? k

We may choose the team members
 first: $\binom{n}{k}$ and then choose
 the captain: $\binom{k}{1} = k$ (LHS)

We may choose the captain first:
 $\binom{n}{1} = n$, then choose the rest of
the team: $\binom{n-1}{k-1}$ (RHS)

Prop 4 $\binom{n}{2} = 1 + 2 + 3 + \dots + (n-1)$

Proof LHS: # of 2-element subsets
of $A = \{1, 2, \dots, n\}$

RHS: Here is a possible breakdown
of the subsets:

There are $n-1$ subsets with "1":

$$\{1, 2\}, \{1, 3\}, \dots, \{1, n\}$$

There are $n-2$ subsets without "1"
but with "2":

$$\{2, 3\}, \{2, 4\}, \dots, \{2, n\}$$

There are $n-3$ subsets w/o "1" or "2"
but with "3": $\{3, 4\}, \{3, 5\}, \dots, \{3, n\}$

⋮
There is 1 subset without "1" -- "n-2"
but w: "n-1": $\{n-1, n\}$

Prop 5 $\sum_{k=0}^l \binom{n}{k} \binom{m}{l-k} = \binom{n+m}{l}$

$\underbrace{\binom{n}{0} \binom{m}{l}}_{k=0} + \underbrace{\binom{n}{1} \binom{m}{l-1}}_{k=1} + \dots$

$\underbrace{\binom{n}{l} \binom{m}{0}}_{k=l}$

choose l marbles from n red marbles and m blue marbles

$0 \text{ red} + l \text{ blue}$

$1 \text{ red} + l-1 \text{ blue}$

$l \text{ red} + 0 \text{ blue}$